

Notes: Olsson & Tåg (JF, 2025)

What Is the Cost of Privatization for Workers?

Contents

1	Overview	3
2	Economic motivation and key intuition	3
2.1	Why worker costs are central in privatization	3
2.2	Empirical prediction	3
3	Data and sample construction	4
3.1	Administrative micro data	4
3.2	Treatment definition: privatization	4
3.3	Matched treated and control groups	4
4	Empirical strategy: stacked DID and event studies	4
4.1	Why “stacked” DID?	4
4.2	Worker-level baseline DID (average post effect)	5
4.3	Worker-level dynamic DID (event-study effects)	5
4.4	Firm-level DID	5
4.5	Job-flow definitions (employment growth, destruction, creation)	6
5	Baseline comparability: Table I	6
5.1	What Table I checks	6
5.2	Interpretation	6
5.3	Table I	6
6	Main worker outcomes: wages, unemployment, transfers (Table II and Fig. 3)	6
6.1	Outcome definitions (Table II)	6
6.2	Average effect (Table II, Panel A)	7
6.3	Dynamic effects (Table II, Panel B; Fig. 3)	7
7	Other worker outcomes (Table III and Fig. 4A–C)	8
7.1	What Table III measures	8
7.2	Interpretation of Table III	9

8	Household finance outcomes (Table IV and Fig. 4D–F)	9
8.1	What Table IV measures	9
8.2	Interpretation of Table IV	9
9	Firm-level outcomes: employment, job flows, profitability, productivity (Table V and Figs. 5–6)	9
9.1	What Table V measures	9
9.2	Interpretation of Table V	10
10	Mechanisms: labor quality, financial engineering, governance (Table VI and Fig. 6)	11
10.1	What Table VI tests	11
10.2	Interpretation of Table VI	11
11	Heterogeneity and special cases (Tables VII–IX)	11
11.1	Partial privatizations (Table VII)	11
11.2	Privatization type: share issue, foreign buyer, financial buyer, MBO (Table VIII) . .	12
11.3	Additional heterogeneity (Table IX)	12
12	Distributional trade-off: worker losses vs productivity gains	13
12.1	A simple accounting logic	13
12.2	Interpretation of the paper’s accounting	13
13	Limitations and “what I would do next” (optional research notes)	14
14	Figures 1–2 (descriptive context)	14
14.1	Figure 1	14
14.2	Figure 2	14
15	Bottom line (the conclusion in plain English)	15
16	Conclusion	15
16.1	Summary	15
16.2	Main empirical takeaways (what changes after privatization)	15
16.3	Mechanism intuition (why productivity rises when workers lose)	16
16.4	Welfare and policy intuition	16

1 Overview

The paper asks a simple but policy-relevant welfare question: *when a state-owned enterprise (SOE) is privatized, what happens to the incumbent workers?* Using Swedish administrative registers that link workers to employers over time, the authors match workers employed in SOEs that are privatized to similar workers employed in SOEs that remain state-owned, and then estimate stacked difference-in-differences (DID) event studies around the privatization year. They document economically meaningful *worker-level costs*—lower wage income and higher unemployment risk—together with higher government transfers that partially smooth income losses. At the same time, privatized firms display sizable *productivity and profitability gains*, largely driven by employment reductions and (especially) by governance changes such as CEO replacement rather than by improvements in worker “talent” or financial engineering. A back-of-the-envelope accounting implies that productivity gains per worker exceed wage losses by a factor of roughly two to six, and that transfers sufficient to fully compensate workers would absorb only a modest share (about 10–30%) of productivity gains.

2 Economic motivation and key intuition

2.1 Why worker costs are central in privatization

A government may keep firms state-owned not only because of “market failures” but also to pursue *non-pecuniary objectives* such as stabilizing employment, protecting certain regions, or providing implicit insurance to workers. If this is true, then privatization should be expected to reduce those implicit benefits:

- **Implicit insurance channel:** SOEs may provide higher job security and wage stability; privatization may increase job loss risk.
- **Soft constraints and slack:** SOEs may tolerate labor hoarding and lower effort; privatization may remove slack and cut labor.
- **Governance channel:** private owners can replace management, redesign incentives, and reallocate resources; incumbent workers can be negatively affected even if firm efficiency rises.

2.2 Empirical prediction

If SOEs provide implicit worker insurance, then after privatization one should observe:

wages ↓, unemployment risk ↑, government transfers ↑,

and, at the firm level,

employment ↓, job destruction ↑, profitability/productivity ↑.

The paper is fundamentally about measuring these objects *causally* and then comparing their magnitudes.

3 Data and sample construction

3.1 Administrative micro data

The analysis uses Swedish administrative registers that allow the authors to follow:

- **Workers:** annual labor income, unemployment status and unemployment days, government transfers, demographics, education, labor market experience, tenure, etc.
- **Firms:** employment, payroll, accounting outcomes (e.g., ROA), value added and productivity, firm age, industry, and region.
- **Households (subset period):** portfolio participation and composition (stock-market participation, risky share, leverage/debt ratio).

3.2 Treatment definition: privatization

A privatization event is a discrete reduction in government ownership/control of an SOE. The analysis distinguishes:

- **Full privatizations** (baseline),
- **Partial privatizations** (separate analysis, important for heterogeneity).

3.3 Matched treated and control groups

Worker-level matched sample. For each privatization event k occurring in year T_k , the authors identify:

- **Treated workers:** employed in the privatized firm one year prior to the privatization (event time 0).
- **Controls:** workers employed one year prior in *similar SOEs that are not privatized* (and that are matched to treated workers).

The matched pairs are created within a set of observable worker and firm characteristics to ensure that treated and control workers look similar prior to the event.

Firm-level matched sample. Similarly, privatized firms are matched to non-privatized SOE firms that are comparable in observed pre-privatization characteristics. Firm-level panels are constructed with a pre-period and a post-period around the privatization date.

4 Empirical strategy: stacked DID and event studies

4.1 Why “stacked” DID?

Privatization is staggered over time. In staggered designs, naive two-way fixed effects can use already-treated units as controls for later-treated units, potentially creating bias when treatment effects are heterogeneous over event time. The stacked DID design instead builds a separate “experiment” around each privatization event k (treated units and matched controls that are not treated in that window) and then stacks these event-specific panels.

4.2 Worker-level baseline DID (average post effect)

For worker i employed at firm f around privatization event k , the paper estimates:

$$Y_{i,f,k,t} = \alpha + \pi \text{After}_{k,t} + \gamma D_i + \beta (\text{After}_{k,t} \times D_i) + \omega_t + X_i + X_f + \varepsilon_{i,f,k,t}, \quad (1)$$

where:

- $D_i = 1$ if worker i is in a privatized firm one year prior to privatization; $D_i = 0$ for the matched control.
- $\text{After}_{k,t} = 1$ for post-privatization years (event time ≥ 1), and 0 otherwise (event time ≤ 0).
- ω_t are calendar-year fixed effects; X_i and X_f are worker and firm controls (measured one year before privatization, except year and privatization-year FE).
- Standard errors are clustered at the municipality level (as stated in the table notes).

Interpretation: β is the DID estimate of the average effect of privatization on outcome Y .

4.3 Worker-level dynamic DID (event-study effects)

To trace dynamics and check pre-trends, the paper estimates:

$$Y_{i,f,k,t} = \alpha + \tau_e + \gamma D_i + \sum_{e \neq 0} \beta_e (\tau_e \times D_i) + \omega_t + X_i + X_f + \varepsilon_{i,f,k,t}, \quad (2)$$

where τ_e are event-time dummies (e.g., $e \in \{-3, -2, -1, 0, 1, \dots, 8\}$) and event time 0 is the year *before* privatization. **Interpretation:** β_e measures the causal difference between treated and control workers at event time e relative to the baseline year $e = 0$. Pre-trend diagnostics correspond to whether β_e is near zero for $e < 0$.

4.4 Firm-level DID

For firm outcomes $Y_{f,k,t}$, the baseline DID is:

$$Y_{f,k,t} = \alpha + \pi \text{After}_{k,t} + \gamma D_f + \beta (\text{After}_{k,t} \times D_f) + \omega_t + X_f + \varepsilon_{f,k,t}, \quad (3)$$

where $D_f = 1$ for privatized firms and 0 for matched non-privatized SOE firms. The firm-level specifications include rich fixed effects (privatization-year, calendar-year, industry, region, firm age) and pre-privatization controls.

4.5 Job-flow definitions (employment growth, destruction, creation)

For firm-level employment and job flows, the paper uses standard Davis–Haltiwanger style measures:

$$g_{f,t} = \frac{E_{f,t} - E_{f,t-1}}{0.5(E_{f,t} + E_{f,t-1})}, \quad (4)$$

$$\text{Destruction}_{f,t} = \frac{\max\{E_{f,t-1} - E_{f,t}, 0\}}{0.5(E_{f,t} + E_{f,t-1})}, \quad (5)$$

$$\text{Creation}_{f,t} = \frac{\max\{E_{f,t} - E_{f,t-1}, 0\}}{0.5(E_{f,t} + E_{f,t-1})}. \quad (6)$$

These definitions are useful because they decompose net employment growth into gross job destruction and creation.

5 Baseline comparability: Table I

5.1 What Table I checks

Table I reports summary statistics for treated and control groups, separately for:

- **Panel A: Workers** (age, gender, immigrant status, education, tenure, pre-privatization wages, unemployment, transfers, etc.).
- **Panel B: Firms** (employment, payroll, ROA, productivity, firm age, etc.).

5.2 Interpretation

The main purpose is a balance check: treated and control units look similar *in the year before privatization* (event time 0). This supports (but does not prove) the key identifying assumption that, absent privatization, treated and control workers/firms would have evolved similarly.

5.3 Table I

6 Main worker outcomes: wages, unemployment, transfers (Table II and Fig. 3)

6.1 Outcome definitions (Table II)

The main worker outcomes are:

- **Wages:** log of annual gross salary income, using the inverse hyperbolic sine (IHS) transform.
- **Unemployment:** indicator equal to one if unemployed at any time during the year.
- **Transfers:** log of annual gross government transfers (unemployment benefits, active support payments, social benefits), using IHS.
- **Income:** log of wage income plus transfers, using IHS.

IHS is used because it is well-defined at zero and behaves like the log for large values; coefficients can be read approximately as percent changes.

Table I
Comparison of Treated and Control Groups

The table displays mean characteristics of treated and control workers and firms one year before the privatization. A normalized t -test above 0.25 indicates substantial differences in means (Imbens and Wooldridge (2009)). *Wage* is total annual wage income in thousands of SEK, *Transfers* is the sum of unemployment benefits, activity support, and social benefits in thousands of SEK, *Total income* is wages plus transfers, *Labor market experience* is the number of years since entering the labor market, and *Tenure* is the number of years employed in the firm. *Employees* is the number of employees, *Payroll* is the sum of the annual wage income of workers employed in the firm in hundreds of SEK, *ROA* is operating return on assets, and *Productivity* is the value-added per employee in thousands of SEK. *Investment ratio* is capital expenditures divided by total assets. *Leverage* is the sum of short- and long-term liabilities divided by total assets.

	Treated (1)	Control (2)	Difference (3)	Norm. t -Value (4)
Panel A: Worker Level				
<i>Female</i>	36%	36%	0%	0.00
<i>Immigrant</i>	15%	13%	2%	0.05
<i>Age</i>				
20–33	30%	30%	0%	0.00
34–43	26%	26%	0%	0.00
44–52	24%	24%	0%	0.00
53–60	20%	20%	0%	0.00
<i>Education</i>				
Basic	15%	12%	3%	0.06
High school	53%	53%	0%	0.00
Vocational	16%	16%	0%	−0.01
University	16%	19%	−2%	−0.05
<i>Wage</i>	271.08	265.72	5.36	0.02
<i>Transfers</i>	4.07	3.77	0.31	0.01
<i>Total income</i>	275.15	269.49	5.69	0.02
<i>Labor market experience</i>				
0–5	18%	18%	−1%	−0.01
6–10	11%	11%	0%	0.00
11–20	25%	25%	0%	0.00
21–30	23%	23%	0%	−0.01
30+	23%	22%	1%	0.02
<i>Tenure</i>				
0–2	59%	56%	4%	0.05
3–5	21%	21%	0%	0.00
6–10	15%	19%	−4%	−0.08
11–15	3%	3%	0%	0.00
16+	2%	2%	0%	0.02
<i>Observations</i>	63,231	63,231		
Panel B: Firm Level				
<i>Employees</i>	107.00	118.90	−11.90	−0.02
<i>Payroll</i>	279.58	302.37	−22.79	−0.02
<i>ROA</i>	0.00	0.02	−0.02	−0.06
<i>Productivity</i>	270.93	308.01	−37.08	−0.08

Table I—Continued

Panel B: Firm Level				
<i>Investment ratio</i>	0.06	0.08	−0.01	−0.08
<i>Leverage</i>	0.63	0.68	−0.05	−0.14
<i>Observations</i>	368	368		

6.2 Average effect (Table II, Panel A)

The average post-privatization effect (equation (1)) implies:

- **Wage income falls:** about -7.9% .
- **Unemployment increases:** about $+1.3$ percentage points (roughly $+12.4\%$ relative to the baseline mean).
- **Transfers rise:** about $+11.9\%$ (consistent with the state smoothing unemployment/wage shocks).
- **Total income declines:** about -3.5% (transfers partially offset wage losses).

Economic reading: privatization reduces the implicit insurance provided by SOEs; the welfare state partially replaces it via transfers.

6.3 Dynamic effects (Table II, Panel B; Fig. 3)

The dynamic estimates (equation (2)) show:

- **Pre-trends are small** (visual check in Fig. 3): coefficients before event time 0 are close to zero.

Table II
Wages, Unemployment, and Government Transfers

The table reports difference-in-differences estimates from regressions explaining worker outcomes after privatization using the models in equations (1) and (2) (*t*-statistics in parentheses). The sample consists of treated workers employed one year prior to the privatization and matched control workers. All models control for the number of employees and fixed effects for *age*, *gender*, *immigrant status*, *labor market experience*, *tenure*, *education*, *municipality*, *industry*, *calendar year*, and *privatization year*. All controls, except calendar and privatization years, are measured one year before the privatization. Standard errors are clustered at the municipality level. %change uses the mean for workers in privatized firms in the years before privatization as the baseline. *Wage* is the log of annual gross salary income (using the inverse hyperbolic sine transformation), *Unemployment* takes the value one if an individual was unemployed at any time during the year and zero otherwise, *Transfers* is the log of annual gross government transfers to workers (unemployment benefits, active support payments, and social benefits (using the inverse hyperbolic sine transformation)), and *Income* is the log of the sum of wages and transfers (using the inverse hyperbolic sine transformation).

Dependent Variable Specification	Wages (1)	Unemployment (2)	Transfers (3)	Income (4)
Panel A: Average Effect				
Full period	-0.079 (-2.96)	0.013 (5.16)	0.119 (5.82)	-0.035 (-1.52)
%-Change	-7.9%	12.4%	11.9%	-3.5%
Adjusted R^2	0.121	0.072	0.064	0.129
Panel B: Dynamic Effect				
Short run (1–2 years)	-0.058 (-3.47)	0.011 (4.02)	0.099 (4.15)	-0.029 (-2.61)
Medium run (3–4 years)	-0.093 (-4.13)	0.012 (4.27)	0.112 (5.27)	-0.043 (-2.15)
Long run (5–8 years)	-0.084 (-2.23)	0.015 (5.27)	0.135 (6.09)	-0.034 (-1.00)
%-Change				
Short run	-5.8%	10.5%	9.9%	-2.9%
Medium run	-9.3%	11.4%	11.2%	-4.3%
Long run	-8.4%	14.3%	13.5%	-3.4%
Adjusted R^2	0.123	0.072	0.065	0.131
Mean dep. var.	7.995	0.105	0.723	8.142
Observations	1,414,270	1,414,270	1,414,270	1,414,270

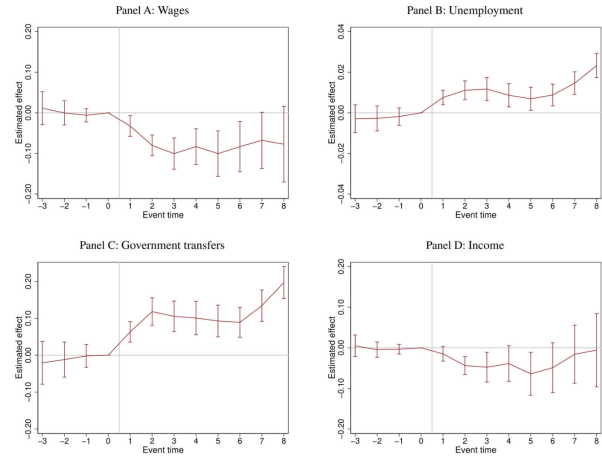


Figure 3. Effects of privatization on wages, unemployment, government transfers, and income. The figures display dynamic difference-in-differences estimates relative to the year prior to the privatization (event time 0) using the model in equation (2). The vertical bars display 95% confidence intervals using robust standard errors clustered at the municipality level. Variable descriptions are available in Table II. (Color figure can be viewed at wileyonlinelibrary.com)

- **Effects grow over time:** wage losses and unemployment increases are larger in the medium and long run than in the short run.
- **Transfers respond more in the long run,** consistent with persistent labor-market disruption for some workers.

7 Other worker outcomes (Table III and Fig. 4A–C)

7.1 What Table III measures

Beyond wage income and unemployment incidence, Table III studies:

- **Unemployment days:** annual total number of unemployment days.
- **Out of labor force:** indicator for no employment, no wages, and no unemployment days (a “non-participation” proxy).
- **Retired:** indicator for receiving retirement income.
- **Self-employed:** indicator for running a sole proprietorship.
- **Business owner:** indicator for being employed in a limited-liability firm that the individual owns.
- **Divorce:** indicator for a transition from married to unmarried.
- **Mortality:** deaths per thousand (estimated using post years only, with a special specification).

7.2 Interpretation of Table III

Key patterns:

- **Unemployment intensity rises materially:** unemployment days increase by about 19.5% on average, and the dynamic pattern in Panel B shows increases that grow in the long run.
- **Limited evidence of “discouragement” or early retirement:** out-of-labor-force and retirement indicators show small, statistically weak movements.
- **Entrepreneurship margin:** business ownership increases (the percent change is large because the baseline mean is small), consistent with some workers switching into ownership/entrepreneurship, but the level effect is modest.
- **Family and health outcomes:** divorce increases slightly; mortality effects are statistically weak.

8 Household finance outcomes (Table IV and Fig. 4D–F)

8.1 What Table IV measures

Using household portfolio data (available for a limited period), the paper studies whether privatization shocks change:

- **Stock-market participation:** indicator for owning risky stocks.
- **Risky share:** share of risky assets in total financial assets.
- **Leverage (debt ratio):** personal debt divided by personal assets.

The analysis restricts to privatizations between 1999 and 2007 and uses shorter windows (two years pre and two years post) given data availability.

8.2 Interpretation of Table IV

The estimates are close to zero:

- No meaningful change in stock-market participation or risky share.
- No meaningful change in leverage.

Reading: the main worker-level response appears in labor-market outcomes rather than in household portfolio re-optimization (at least at annual horizons and for these measures).

9 Firm-level outcomes: employment, job flows, profitability, productivity (Table V and Figs. 5–6)

9.1 What Table V measures

Firm outcomes include:

Table IV
Household Finance Outcomes

The table reports difference-in-differences estimates from regressions explaining worker outcomes after privatization using the models in equations (1) and (2) (*t*-statistics in parentheses). See Table II for details on the sample and control variables. *Stock market participation* takes the value one if an individual owns risky stocks, *Risky share* measures the share of risky assets to total assets of an individual's portfolio, and *Leverage* measures total personal debt to total personal assets.

Dependent Variable Specification	Participation (1)	Risky share (2)	Debt ratio (3)
Panel A: Average Effect			
Full period	-0.0005 (-0.15)	-0.0032 (-0.88)	-0.0002 (-0.05)
%-Change	-0.1%	-0.7%	0.0%
Adjusted R^2	0.045	0.032	0.108
Panel B: Dynamic Effect			
Short run	0.0016 (0.48)	-0.0049 (-1.64)	-0.0010 (-0.24)
Medium run	-0.0027 (-0.67)	-0.0015 (-0.32)	0.0006 (0.12)
%-Change			
Short run	0.3%	-1.0%	-0.2%
Medium run	-0.4%	-0.3%	0.1%
Adjusted R^2	0.045	0.032	0.108
Mean dep. var.	0.639	0.470	0.560
Observations	342,554	342,554	342,554

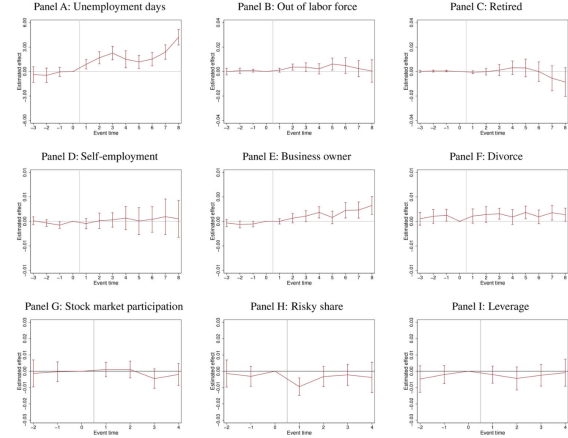


Figure 4. Effects of privatization on other worker outcomes. The figures display dynamic difference-in-differences estimates relative to the year prior to the privatization (event time 0) using the model in equation (2). The vertical bars display 95% confidence intervals using robust standard errors clustered at the municipality level. Variable descriptions are available in Tables III and IV. (Color figure can be viewed at wileyonlinelibrary.com)

- **Employees and Payroll** (labor scale).
- **Job destruction and job creation** rates (equations (5)–(6)).
- **ROA** (operating return on assets).
- **Productivity** (value added per employee).

Specifications include fixed effects for privatization year, calendar year, industry, region, and firm age; controls are measured one year before privatization.

Table V
Firm-Level Outcomes

The table reports difference-in-differences estimates from regressions explaining worker outcomes after privatization using the models in equation (3) adjusted to account for dynamic effects (*t*-statistics in parentheses). The sample consists of treated firms and matched control firms and includes information for three years before and four years after the privatization. All models include fixed effects for privatization year, calendar year, industry, region, and firm age. All controls, except calendar and privatization years, are measured one year before the privatization. %-change uses the mean for privatized firms over the years before the privatization as the baseline. *Employees* is the number of employees, *Destruction (Creation)* measures the job destruction and creation rates as defined in equations (5) and (6). *Payroll* is the sum of the annual wages income of workers employed in the firm in hundreds of SEK, *ROA* is operating return on assets, and *Productivity* is the value added per employee in thousands of SEK.

Dependent Variable Specification	Employees (1)	Destruction (2)	Creation (3)	Payroll (4)	ROA (5)	Productivity (6)
Panel A: Average Effect						
Full period	-0.163 (-2.82)	0.109 (4.92)	-0.025 (-1.40)	-0.122 (-2.13)	0.021 (1.82)	109.454 (2.77)
%-Change	-16.3%	10.9%	-2.5%	-12.2%	321.3%	35.7%
Adjusted R^2	0.186	0.448	0.721	0.224	0.074	0.072
Panel B: Dynamic Effect						
Short run (1–3 years)	-0.141 (-2.81)	0.118 (4.40)	-0.027 (1.49)	-0.124 (-2.45)	0.018 (1.46)	101.936 (2.91)
Medium run (3–4 years)	-0.196 (-2.32)	0.098 (3.22)	-0.022 (-1.06)	-0.122 (-1.45)	0.026 (1.69)	122.514 (2.12)
%-Change						
Short run	-14.1%	11.8%	-2.7%	-12.4%	273.8%	33.3%
Medium run	-19.6%	9.8%	-2.2%	-12.2%	392.1%	39.9%
Adjusted R^2	0.186	0.448	0.721	0.224	0.074	0.072
Mean dep. var.	0.196	0.072	0.375	11.392	0.007	206.655
Observations	4,804	4,611	4,611	4,804	4,804	4,804

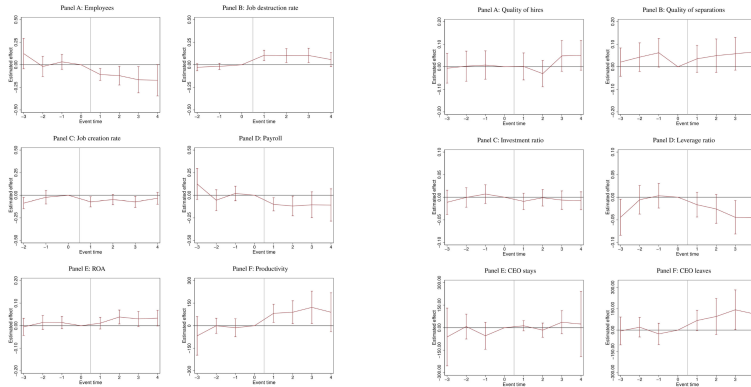


Figure 5. Effects of privatization on firm-level outcomes. The figures display dynamic difference-in-differences estimates relative to the year prior to the privatization (event time 0) using the model in equation (3). The vertical bars display 95% confidence intervals using robust standard errors clustered at the municipality level. Variable descriptions are available in Tables III and IV. (Color figure can be viewed at wileyonlinelibrary.com)

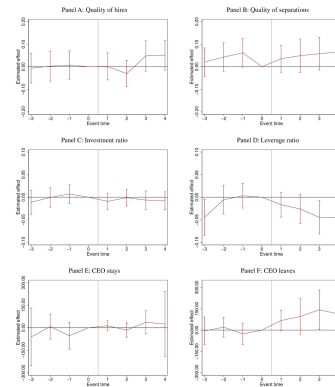


Figure 6. Effects on additional firm-level outcomes. The figures display dynamic difference-in-differences estimates relative to the year prior to the privatization (event time 0) using the model in equation (3). The vertical bars display 95% confidence intervals using robust standard errors clustered at the municipality level. Variable descriptions are available in Tables III and IV. (Color figure can be viewed at wileyonlinelibrary.com)

9.2 Interpretation of Table V

Main patterns:

- **Employment scale shrinks:** employees drop by about 16% and payroll by about 12%.
- **Reallocation occurs through destruction:** job destruction increases by about 11%, while job creation does not increase (slightly negative and statistically weak).
- **Profitability increases:** ROA rises substantially relative to a small pre-mean.

- **Productivity rises:** value added per employee rises strongly (the table reports large percentage increases).

Interpretation: the firm-level productivity improvement is tightly linked to labor shedding and restructuring rather than to expansion.

10 Mechanisms: labor quality, financial engineering, governance (Table VI and Fig. 6)

10.1 What Table VI tests

Table VI examines candidate channels behind productivity gains:

- **Labor quality:** “Quality of hires” and “quality of separations” measured using wage residuals from an auxiliary regression with rich controls (a higher residual indicates higher worker quality).
- **Financial engineering:** *Investment ratio* (investments to lagged assets) and *leverage ratio* (debt to assets).
- **Governance change:** compare productivity gains when the CEO stays versus when the CEO is replaced in the post-privatization period.

10.2 Interpretation of Table VI

The evidence points to governance rather than compositional labor upgrading or financial leverage:

- **No strong change in worker “talent”:** quality of hires and separations are close to zero.
- **No evidence of large investment boom or leverage increase:** investment ratio changes are small; leverage ratio declines.
- **CEO replacement matters:** productivity gains are much larger when the CEO is replaced than when the CEO stays, consistent with a governance engineering channel (new owners actively change management).

11 Heterogeneity and special cases (Tables VII–IX)

11.1 Partial privatizations (Table VII)

Table VII focuses on *partial* privatizations. Compared with the baseline full-privatization results, the costs for workers are larger:

- larger wage losses,
- larger increases in unemployment,
- and larger declines in total income.

Table VI
Additional Firm-Level Outcomes

The table reports difference-in-differences estimates from regressions explaining firm outcomes after privatization using the model in equation (3) adjusted to account for dynamic effects (*t*-statistics in parentheses). See Table V for details on the sample and control variables. *Quality Hir* is the mean talent of hired persons relative to the mean talent of the incumbent workforce, where talent is defined as the sum of cognitive and noncognitive ability measures at military enlistment for males. *Quality Sep* is the mean talent of persons leaving the firm relative to the mean talent of the incumbent workforce. *Investment ratio* is capital expenditures divided by total assets. *Leverage* is the sum of short- and long-term liabilities divided by total assets. *Productivity* is the value-added per employee in thousands of SEK, and *CEO remains* refers to the subsample of privatizations in which the CEO is not replaced in the postperiod. Specifications (1) to (4) use the full sample while the sample in specification (5) includes all control firms but only treated firms in which the CEO remained the same during the postperiod. The sample in specification (6) includes all control firms but only treated firms in which the CEO was replaced during the postperiod. Standard errors are clustered at the firm level.

Dependent Variable	<i>Qual. Hir</i>	<i>Qual. Sep</i>	<i>Investment r.</i>	<i>Leverage</i>	<i>Prod. CEO stays</i>	<i>Prod. CEO leaves</i>
Sample Specification	Full (1)	Full (2)	Full (3)	Full (4)	(5)	(6)
Panel A: Average Effect						
Full period	0.012 (0.72)	0.020 (1.15)	-0.006 (-0.53)	-0.022 (-1.81)	39.942 (0.81)	97.875 (2.00)
%-Change	1.2%	1.9%	-8.4%	-3.5%	12.8%	31.4%
Adjusted R^2	0.004	0.025	0.063	0.108	0.116	0.077
Panel B: Dynamic Effect						
Short run (1–2 years)	-0.015 (-0.78)	0.011 (0.52)	-0.005 (-0.76)	-0.012 (-0.91)	26.624 (0.64)	81.786 (1.82)
Medium run (3–4 years)	0.048 (2.12)	0.031 (1.41)	-0.007 (-0.91)	-0.036 (-2.02)	58.083 (0.77)	130.071 (1.65)
%-Change						
Short run	-1.5%	1.1%	-7.2%	-1.9%	8.5%	26.2%
Medium run	4.8%	3.0%	-9.9%	-5.7%	18.6%	41.7%
Adjusted R^2	0.006	0.026	0.063	0.108	0.116	0.077
Mean dep. var.	1.011	1.012	0.069	0.630	311.749	311.749
Observations	2,957	2,757	4,804	4,804	2,433	2,278

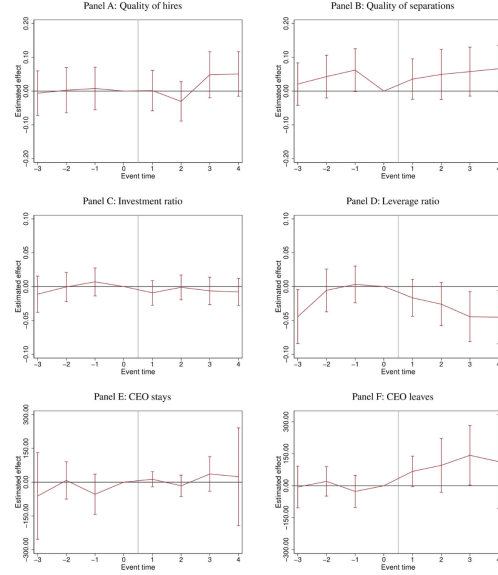


Figure 6. Effects on additional firm-level outcomes. The figures display dynamic difference-

Interpretation: partial privatizations (often associated with listing/share issues) may introduce market discipline and restructuring pressures while still leaving some features of the SOE environment, producing especially pronounced worker disruption in this setting.

11.2 Privatization type: share issue, foreign buyer, financial buyer, MBO (Table VIII)

Table VIII shows that worker impacts vary with buyer type:

- **Share-issue/stock-market listing:** larger unemployment increases; income effects differ across listed vs non-listed cases.
- **Foreign buyers:** much larger negative income effects and higher unemployment than domestic buyers.
- **Financial buyers:** do not stand out relative to others on these outcomes in a sharp way (in the reported averages).
- **Management buyouts (MBO):** effects are relatively benign (unemployment may even fall, though estimates are noisy).

Reading: ownership identity matters for worker welfare, consistent with different reorganization and cost-cutting strategies.

11.3 Additional heterogeneity (Table IX)

Table IX runs triple-difference regressions to test whether worker impacts differ:

- across specific industries (transport and business activity/intermediary),
- across local labor markets (high vs low unemployment regions),

- and across macro conditions (recession vs non-recession years).

The reported DDD effects show limited systematic heterogeneity across these splits.

Table VII
Wages, Unemployment, and Government Transfers for Partial Privatizations

The table reports difference-in-differences estimates from regressions explaining worker outcomes after privatization using the models in equations (1) and (2) (t-statistics in parentheses). The sample consists of treated workers employed one year prior to the partial privatization and matched control workers. See Table I for details on the variables and the controls used in the regressions.

Dependent Variable Specification	Wage (1)	Unemployment (2)	Transfers (3)	Total Income (4)
Panel A: Average Effect				
Full period	-0.165 (-4.11)	0.025 (0.77)	0.167 (6.19)	-0.109 (-3.25)
%-Change	-16.5%	41.7%	16.7%	-30.9%
Adjusted R ²	0.137	0.062	0.051	0.150
Panel B: Dynamic Effect				
Short run (1-2 years)	-0.061 (-2.77)	-0.000 (-0.10)	0.006 (0.29)	-0.027 (-2.15)
Medium run (3-4 years)	-0.090 (-2.81)	0.025 (6.50)	0.134 (5.59)	-0.073 (-2.57)
Long run (5-8 years)	-0.255 (-4.56)	0.037 (6.66)	0.266 (7.47)	-0.163 (-3.48)
%-Change	-6.1%	0.5%	0.6%	-3.7%
Short run	-9.6%	41.7%	13.6%	-7.3%
Medium run	-25.5%	61.7%	26.6%	-16.3%
Long run	0.137	0.062	0.050	0.150
Adjusted R ²	0.066	0.060	0.387	8.166
Mean dep. var.	681.113	681.113	681.113	681.113
Observations	681,113	681,113	681,113	681,113

Table VIII
Heterogeneity across Privatization Type

The table reports difference-in-differences estimates from regressions explaining worker outcomes after privatization using the model in equation (1) (t-statistics in parentheses). Each subsample consists of treated workers employed one year prior to privatization and matched control workers. See Table I for details on the variables and the controls used in the regressions. *Share issue* is the sample of treated workers where the privatized firm (or its new owner) is listed on the stock market, *Foreign buyer* refers to the sample of treated workers where the new owner of the privatized firm is a foreign firm, *Financial buyer* refers to new owners with industry codes corresponding to "Investment trust activities," "Security brokering and fund management," "Activities auxiliary to financial intermediation," or "Activities of investment companies and venture capital companies," and *MBO* refers to privatizations in which at least one of the management/employees at $t-1$ is the new owner, that is, the firm is privatized through a management buyout.

Type Specification	Share issue		Foreign buyer		Financial buyer		MBO	
	Yes	No	Yes	No	Yes	No	Yes	No
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Unemployment								
Full period	0.021 (7.15)	0.010 (3.27)	0.032 (8.59)	0.005 (1.86)	0.010 (1.36)	0.013 (4.95)	-0.020 (-0.96)	0.013 (5.21)
%-Change	22.8%	10.9%	32.0%	5.0%	10.2%	13.3%	-20.7%	13.9%
%-Value for diff.	0.008	0.008	0.000	0.000	0.000	0.000	0.000	0.000
Mean dep. var.	0.092	0.092	0.100	0.100	0.098	0.098	0.097	0.097
Adjusted R ²	0.075	0.075	0.075	0.075	0.083	0.071	0.081	0.072
Panel B: Total Income								
Full period	0.041 (1.56)	-0.060 (-2.51)	-0.166 (-6.07)	0.021 (0.90)	0.034 (0.80)	-0.039 (-1.76)	0.073 (0.54)	-0.036 (-1.54)
%-Change	4.1%	-6.0%	-16.6%	2.1%	3.4%	-3.9%	7.3%	-3.6%
%-Value for diff.	0.000	0.000	0.000	0.000	0.022	0.022	0.435	0.435
Mean dep. var.	8.164	8.164	8.113	8.113	8.126	8.126	8.113	8.113
Adjusted R ²	0.136	0.129	0.135	0.135	0.129	0.129	0.134	0.139
Observations	888,859	1,233,108	912,620	1,209,347	759,558	1,362,409	713,262	1,408,705

Table IX
Heterogeneity across Industries, Labor Markets, and Macroeconomic Conditions

The table reports the triple-difference coefficient from regressions explaining worker outcomes after privatization (t-statistics in parentheses). Each subsample consists of treated workers employed one year prior to privatization and matched control workers. See Table I for details on the variables and the controls used in the regressions. *TB* takes the value one if treated and control workers were employed at time $t-1$ in the transport industry or the business activity/intermediary industry and zero for all other industries. *High unemp.* takes the value one if treated and control workers were employed at time $t-1$ in a region with an above-median unemployment rate. *Recession* takes the value one if there was at least one quarter of negative GDP growth during year $t-1$ and zero otherwise.

Dependent Variable Specification	Unemployment			Total Income		
	TB	High unemp.	Recession	TB	High unemp.	Recession
	(1)	(2)	(3)	(4)	(5)	(6)
DDD	-0.003 (-0.30)	0.011 (1.80)	-0.000 (-0.05)	-0.045 (-1.23)	-0.010 (-0.26)	-0.015 (-0.45)
%-Change	-2.8%	8.6%	-0.3%	-4.5%	-1.0%	-1.5%
Adjusted R ²	0.072	0.072	0.072	0.129	0.129	0.129
Mean dep. var.	0.091	0.134	0.091	8.227	8.107	8.211
Observations	1,414,270	1,414,270	1,414,270	1,414,270	1,414,270	1,414,270

12 Distributional trade-off: worker losses vs productivity gains

12.1 A simple accounting logic

The paper motivates a welfare comparison by contrasting:

- **Worker costs:** wage and income losses, higher unemployment risk, and associated social insurance payments.
- **Efficiency gains:** increases in profitability and productivity at the firm level.

A useful way to organize this is per-worker:

Efficiency gain per worker versus Income loss per worker.

12.2 Interpretation of the paper's accounting

Using the reported magnitudes:

- wage income decreases on the order of 8% on average (larger in some subsamples),
- productivity increases are substantial (and can be interpreted as gains in value added per worker),
- transfers rise, and the implied transfer cost required to fully compensate workers is *small relative to productivity gains*.

The paper emphasizes that productivity gains outweigh wage losses by a factor of roughly *two to six*, and that transfers on the order of *10–30% of productivity gains* could, in principle, compensate workers for their private losses while preserving a large share of the efficiency gains.

13 Limitations and “what I would do next” (optional research notes)

- **External validity:** Sweden has a large welfare state and strong labor institutions; worker costs and transfer responses may differ in economies with weaker social insurance.
- **General equilibrium:** the paper is partial equilibrium in the sense that it focuses on incumbent workers and treated firms; broader effects on local labor markets, competitors, or consumers are not directly quantified.
- **Welfare aggregation:** the accounting contrasts worker losses with firm-level efficiency gains but does not fully value consumer surplus, government revenue, or owner surplus in a unified welfare objective.
- **Mechanisms:** CEO replacement correlates with larger productivity gains, consistent with governance effects; further work could try to disentangle which governance tools (contracts, boards, monitoring, restructuring plans) matter most.

14 Figures 1–2 (descriptive context)

14.1 Figure 1

Figure 1 provides cross-country context, plotting the SOE employment share against social expenditures (as a fraction of GDP). The visual suggests that SOEs and the welfare state can be linked as alternative (or complementary) ways to deliver social objectives.

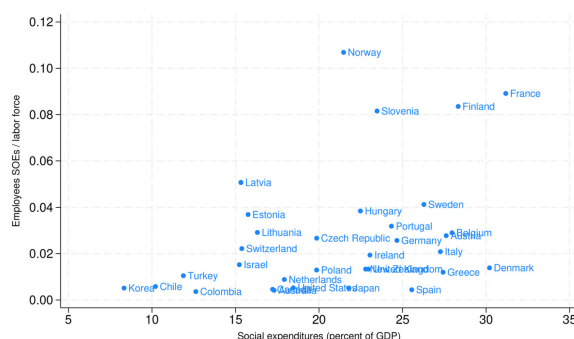


Figure 1. SOE size and government social expenditures. This figure plots the relationship

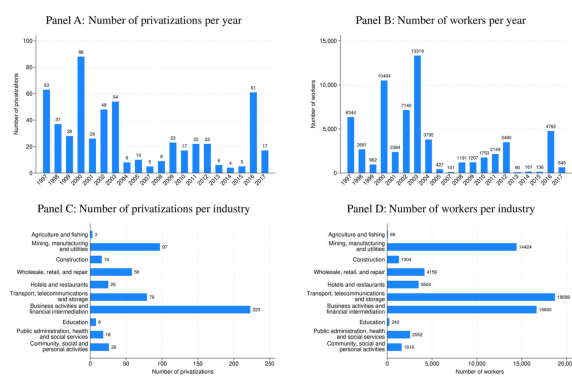


Figure 2. Privatizations vary over time and across industries. The sample consists of privatized firms in Sweden between 1997 and 2017. The number of workers refers to the number of employees in the privatized firms one year before the privatization. The industry classification is based on NACE Rev 1.1. (Color figure can be viewed at wileyonlinelibrary.com)

14.2 Figure 2

Figure 2 summarizes the timing and industry distribution of privatizations and affected workers, highlighting substantial variation across years and sectors, which supports the stacked-event research design.

15 Bottom line (the conclusion in plain English)

1. Privatization creates **real, measurable costs for incumbent workers**: wage income falls and unemployment risk rises.
2. The welfare state **partially offsets** these losses through higher transfers, reducing the decline in total income.
3. Privatized firms experience **large efficiency gains** (higher productivity and profitability) while reducing employment, with job reallocation happening mainly through increased job destruction rather than job creation.
4. Mechanism evidence points away from “hiring better workers” or leverage-fueled investment and toward **governance-driven restructuring**, especially CEO replacement.
5. A simple accounting suggests that the **efficiency gains are large enough** that compensating workers through transfers would use only a modest share of productivity gains.

16 Conclusion

16.1 Summary

The paper concludes that privatization of state-owned enterprises generates a sharp distributional trade-off: *firms become more efficient, but incumbent workers bear meaningful private costs*. Using Swedish linked employer–employee registers and stacked DID event studies around privatization years, the authors find that privatization raises firm productivity and profitability while shrinking employment, and that these firm-level gains are accompanied by lower wage income and higher unemployment for workers who were employed before privatization. The welfare state partially insures workers through higher transfers, so total income falls by less than wage income. Mechanism evidence points primarily to *governance-driven restructuring* (especially CEO replacement) rather than to labor-quality upgrading or leverage-fueled investment. A simple accounting suggests that (measured) productivity gains per worker outweigh workers’ income losses by a factor of roughly two to six, implying that compensating workers fully would require only a modest share of the efficiency gains.

16.2 Main empirical takeaways (what changes after privatization)

1. **Incumbent-worker labor outcomes deteriorate.** On average, wage income declines by about 7.9% while unemployment incidence rises by about 1.3 percentage points (about 12.4% relative to the pre-mean). Unemployment intensity also rises: unemployment days increase by about 19.5%.
2. **Transfers partially offset labor-income losses (social insurance matters).** Government transfers increase by about 11.9%. As a result, *total income* (wages + transfers) falls by only about 3.5%. This pattern implies that a substantial share (roughly half) of the wage-income loss is absorbed by the public safety net.

3. **Non-labor outcomes move little on average, with a few small margins.** The evidence shows no meaningful effects on broad household-finance outcomes such as stock-market participation, risky-asset share, or household leverage. However, in supplementary analyses the authors find an increase in *cash holdings* (a precautionary-savings interpretation is natural), and they find small positive effects on entrepreneurship/business ownership (the proportional change is large because the baseline rate is very low). Family and health outcomes (e.g., mortality) do not show economically large systematic effects in the main analysis.
4. **Firm outcomes improve while employment contracts.** Privatized firms reduce employment (about 16.3% fewer employees) and payroll (about 12.2%). Job reallocation operates mainly via *higher job destruction* (about 10.9%), with no comparable increase in job creation. Profitability rises strongly (ROA increases markedly from a low baseline). Measured labor productivity (value added per worker) increases substantially; across specifications and robustness checks, the paper indicates gains in the ballpark of about 11.5% to 35.7%.

16.3 Mechanism intuition (why productivity rises when workers lose)

A coherent interpretation of the evidence is:

1. **SOEs provide implicit worker insurance and tolerate slack.** State ownership can implicitly insure workers (higher job stability, less aggressive downsizing) and may allow managerial or operational slack.
2. **Privatization changes objectives and governance.** New owners place higher weight on profitability and efficiency, and they can implement governance changes that are harder under state ownership.
3. **Restructuring raises productivity largely by reducing labor input and improving operations.** The paper finds that employment contracts and job destruction rises, consistent with labor shedding and reorganization. At the same time, measured productivity and profitability increase.
4. **Governance (CEO replacement) is a central channel.** Productivity gains are much larger when the CEO is replaced after privatization than when the CEO remains, while changes in labor quality and financial engineering (investment booms or leverage increases) are not the dominant patterns.

16.4 Welfare and policy intuition

- **Efficiency vs. distribution:** Privatization can be efficiency-enhancing at the firm level, but it is not “free” because incumbent workers face earnings losses and higher unemployment risk.
- **Insurance capacity matters:** Because Sweden has a strong welfare state, transfers cushion a large portion of worker losses. In economies with weaker social insurance, the same restructuring could translate into larger uncompensated welfare costs.

- **A feasible compensation logic:** Since measured productivity gains per worker exceed worker income losses by a multiple (roughly 2–6), there is (in principle) fiscal space to design transfer/adjustment policies that compensate affected workers while preserving part of the efficiency gains.
- **A governance takeaway:** The link between CEO turnover and productivity improvements suggests that the “real” gains from privatization may come from changing control, monitoring, and managerial incentives—not simply from changing the financing mix or hiring a different workforce.